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Exercise 1. Among these assertions, specify which ones are true and which ones are false.

- 1- The sample mean is always higher than the population mean.
- 2- The median is always higher than the mean.
- 3- The mean of (1, 2, 8, 9) is 10.
- 4- The STD of a sample is always less than 1.
- 5- The STD of a sample can be -1.
- 6- When median > mean then the distribution is symmetrical.
- 7- Increasing all values of a sample by 5, will increase the mean by 5.
- 8- Increasing all values of a sample by 5, will increase STD by 5.
- 9- Multiplying all values by 2, will multiply the mean by 2.
- 10- Multiplying all values by 2, will multiply the STD by 2.
- 11- The mean is at the middle of the histogram.
- 12- There is 50% chance that one random sample will be in the IQR. ($IQR = Q_3 - Q_1$)
- 13- We can learn the IQR by looking at the boxplot.
- 14- If a value is not in the IQR, then it is an outlier.
- 15- If a value is higher than the mean then it is an outlier.
- 16- Removing an outlier that is positive, will increase the mean.
- 17- Removing an outlier that is negative, will most likely decrease the STD.
- 18- A histogram of any distribution is bell-shaped.
- 19- On the x-axis, a histogram has always positive values.
- 20- The mean is equal to 0 for a symmetric histogram.

[illegible]

Exercise 2. We took note of the performance bonuses (in DA) of the employees of a company. The results are given in the following statistical table :

Bonuses	frequency	Rel. Frequency	Cum. Frequency	Cum. Rel. Frequency
[750, 1250[	26			
[1250, 1750[			97	
[1750, 2250[				0.7
[2250, 2750[			166	
[2750, 3250[				
[3250, 3750[		0.05		
[3750, 4250[	6	0.03		

1. Complete the statistical table above.
2. What is the bonus value received by 50% of the employees.

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3. Give Q_1, Q_3 and interquartile range

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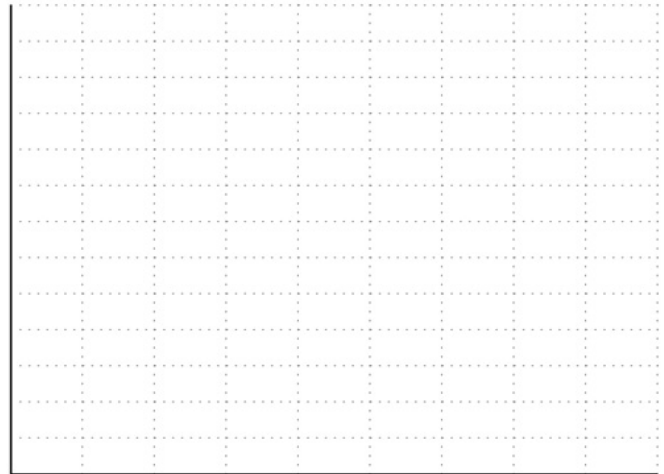
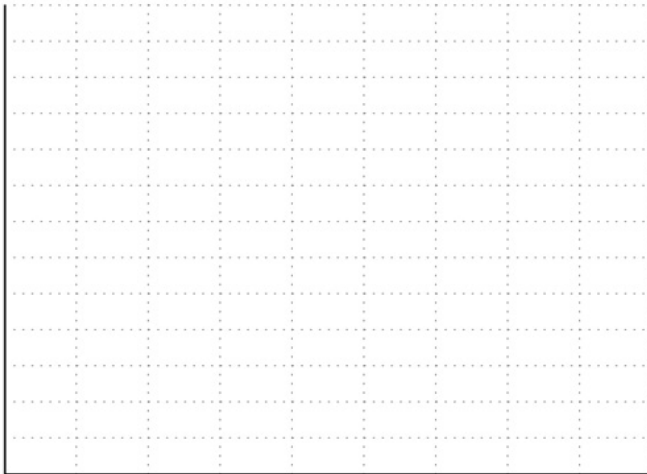
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4. Plot the histogram and the Ogive plot of the above distribution then determine the mode and median of the series graphically



5. Give the modal class then calculate the mode value.

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6. Calculate the mean, variance, standard deviation and coefficient of variation.

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7. Do you think the distribution of the performance bonuses is symmetric? justify.

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Exercise 3. Wheat production (X in quintals) and the number of rainy days (Y) were recorded. The following results were obtained:

[illegible]

1. Draw the scatter plot of the given data. Describe the relationship between X and Y .

[illegible]

2. Calculate the correlation coefficient and interpret the obtained result.

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Show that :

$$\sum_{i=1}^n f_i(x_i - \bar{X})^2 = \min_{a \in \mathbb{R}} \sum_{i=1}^n f_i(x_i - a)^2$$

[illegible]